Area and surface area

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 948 to the nearest hundred.

2. Round 9.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

6. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

7. Find the surface area of a closed 2 by 3 by 5 cm box.

8. Each pack has b beads. Write the total in 8 packs. Find it when b = 4.

9. Expand 3(x + 6).

10. Find 2□. Write the repeated factors.

11. Solve 6x = 54.

12. Use y = 4x for x = 0, 2, 3. Give the ordered pairs.

13. A prism has right-triangle ends with sides 3, 4, 5 cm and length 9 cm. Find surface area.

14. Ava finds a cube’s surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

15. A rectangle has length 11 cm and width 4 cm. Find area and perimeter.

16. A parallelogram has base 9 cm, height 4 cm, and slanted side 6 cm. Find area and name the unused length.

17. A right triangle has perpendicular legs 18 cm and 4 cm. Find area using either leg as base.

18. A triangle has area 36 square cm and base 18 cm. Find its matching height.

CHALLENGE / Explain your plan and check the result

C1. A rectangular sign is 9 cm by 7 cm. A triangle of base 6 cm and perpendicular height 4 cm is cut away. The remaining sign is copied twice more. Find total remaining area for all three copies. Explain a way to calculate without repeating every subtraction.

C2. A cube has side 4 cm. One student counts only the four side faces; another counts all six. Find both totals and the difference. The cube is then opened by removing its top. Which total describes the open box?